

第十五章 正态分布完整教材

(Complete Textbook on Normal Distribution)

第一章 引言:从离散到连续——认识连续型随机变量

(Chapter 1 Introduction: From Discrete to Continuous – Understanding Continuous Random Variables)

现实中随机变量分为两类:离散型(discrete)与连续型(continuous),二者核心特征对比如下:

特征 (Feature)	离散型随机变量 (Discrete Random Variable)	连续型随机变量 (Continuous Random Variable)
取值形式 (Form of Values)	有限/可列离散点(如 1,2,3) (Finite/countable discrete points, e.g., 1,2,3)	充满某区间/实轴(如[1.5,2.8]) (Fills an interval/real axis, e.g., [1.5,2.8])
单个取值概率 (Probability of Single Value)	$P(X = x_0) > 0$ (如掷骰子得 3 的概率 1/6) (e.g., probability of rolling a 3 on a die is 1/6)	$P(X = x_0) = 0$
概率表示方式 (Representation of Probability)	分布列(列举每个取值的概率) (Probability mass function, listing probabilities of each value)	密度函数+积分(区间面积表示概率) (Probability density function + integral, probability represented by interval area)
典型例子(Typical Examples)	投篮命中次数、产品次品数(Number of successful shots, number of defective products)	身高、体重、测量误差、考试成绩(Height, weight, measurement error, exam scores)

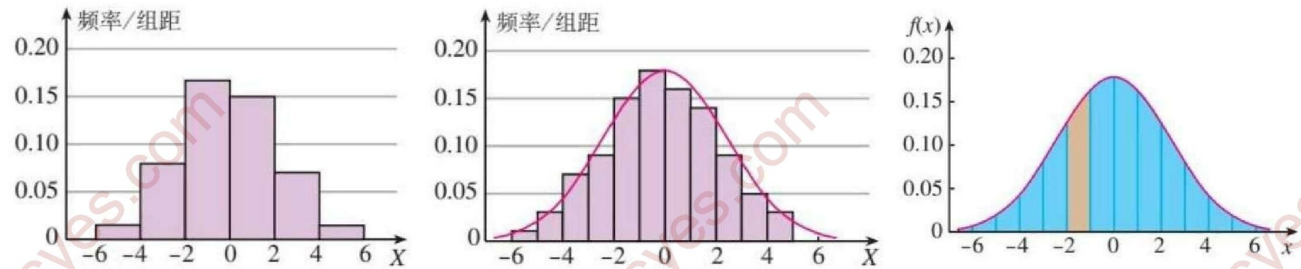
案例:食盐包装的质量误差(Case: Quality Error of Salt Packaging)

自动流水线包装食盐,每袋标准质量 400g,用 X 表示“实际质量-标准质量”的误差(error between actual mass and standard mass)。随机抽取 100 袋,误差数据的频率分布表(frequency distribution table)如下:

误差区间(g) (Error Interval (g))	[-6,-5]	[-5,-4]	[-4,-3]	[-3,-2]	[-2,-1]	[-1,0]	[0,1]	[1,2]	[2,3]	[3,4]	[4,5]	[5,6]
频数(Frequency)	1	2	3	8	15	20	18	12	7	4	1	1
频率/组距 (Frequency/Class Interval)	0.01	0.02	0.03	0.08	0.15	0.20	0.18	0.12	0.07	0.04	0.01	0.01

从表中可直观看出:

- 误差在 $[-1,1]$ 区间的频数最高(38),远离 0 的区间频数快速下降
(The frequency of errors in the interval $[-1,1]$ is the highest (38), and the frequency decreases rapidly for intervals far from 0);
- 正负误差的频数大致对称
(The frequencies of positive and negative errors are roughly symmetric);
- 当样本量扩大到 10000 袋、组距缩小到 0.1 时,频率分布直方图的轮廓会贴合一条钟形曲线(bell-shaped curve)——这就是正态曲线(normal curve)。



第二章 正态分布的定义与数学表达

(Chapter 2 Definition and Mathematical Expression of Normal Distribution)

当连续型随机变量的频率分布直方图趋近于“钟形曲线”时，该变量服从**正态分布**(normal distribution)。

2.1 正态密度函数(Normal Probability Density Function)

正态分布的概率密度函数为(The probability density function of a normal distribution is):

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, \quad x \in \mathbb{R}$$

其中(where):

- $\mu \in \mathbb{R}$: **均值参数**(mean parameter, 决定曲线的中心位置, determines the central position of the curve);
- $\sigma > 0$: **标准差参数**(standard deviation parameter, 决定曲线的“肥瘦”程度, determines the “fatness/leanness” of the curve);
- 分母 $\sigma\sqrt{2\pi}$: **归一化常数**(normalization constant), 保证曲线与 x 轴围成的面积为 1(ensures the area under the curve and above the x -axis is 1, 满足概率分布的基本要求);
- 指数部分(exponential term): x 偏离 μ 越远, $f(x)$ 越趋近于 0, 体现“中间集中、两端稀疏”的特征(the farther x deviates from μ , the closer $f(x)$ is to 0, reflecting the characteristic of “concentration in the middle and sparseness at both ends”).

2.2 正态分布的表示(Notation of Normal Distribution)

若随机变量 X 的密度函数为上述 $f(x)$, 则称 X 服从正态分布, 记为(If the density function of a random variable X is the above $f(x)$, then X is said to follow a normal distribution, denoted as):

$$X \sim N(\mu, \sigma^2)$$

⚠ **易错点**(Common Mistake):第二个参数是**方差**(variance, σ^2), 而非**标准差**(standard deviation, σ)。例如“均值 170、标准差 5”应记为 $N(170, 25)$, 而非 $N(170, 5)$ (For example, “mean 170, standard deviation 5” should be denoted as $N(170, 25)$, not $N(170, 5)$)。

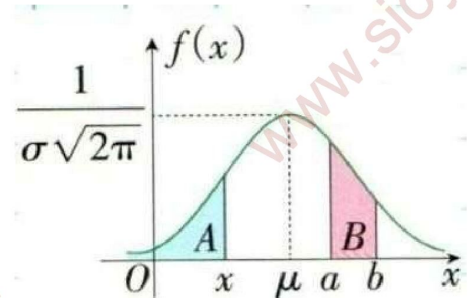
2.3 标准正态分布(Standard Normal Distribution)

当 $\mu = 0$ 、 $\sigma = 1$ 时, 正态分布称为**标准正态分布**(standard normal distribution), 记为 $X \sim N(0, 1)$, 其密度函数简化为(whose density function simplifies to):

$$\varphi(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$$

2.4 正态分布的历史背景(Historical Background of Normal Distribution)

- 1733 年, **棣莫弗**(A. De Moivre)在《机会论》(The Doctrine of Chances)中提出:二项分布 $B(n, p)$ 当 n 很大时, 可近似用“钟形曲线”计算概率, 这是正态分布的雏形(proposed that when the number of trials n of a binomial distribution $B(n, p)$ is large, probabilities can be approximated by a “bell-shaped curve”, which is the prototype of the normal distribution);
- 1809 年, **高斯**(C. F. Gauss)在《天体运动理论》(Theoria Motus Corporum Coelestium)中证明:天体测量误差服从正态分布, 并将其应用于数据修正, 因此正态分布也称为“高斯分布”(Gaussian distribution)。



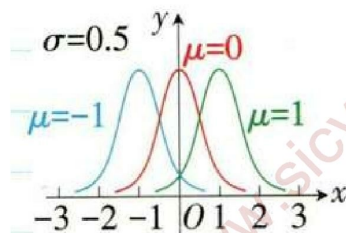
第三章正态曲线的核心特征

(Chapter 3 Core Characteristics of Normal Curve)

正态曲线($f(x)$) 的图像, graph of $f(x)$)具有以下 5 个核心特征(has the following 5 core characteristics):

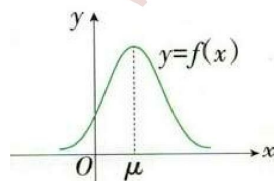
1. **单峰且对称(Unimodal and Symmetric):**

- 曲线关于直线 $x = \mu$ 对称(The curve is symmetric about the line $x = \mu$);
- 实际意义(Practical Significance):若身高 $X \sim N(170,25)$, 则“身高<160cm”的概率等于“身高>180cm”的概率(if height $X \sim N(170,25)$, the probability of “height < 160cm” is equal to the probability of “height > 180cm”).



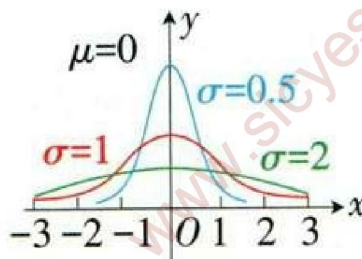
2. **峰值在 $x = \mu$ 处(Peak at $x = \mu$):**

- 峰值为 $f(\mu) = \frac{1}{\sigma\sqrt{2\pi}}$ (The peak value is $f(\mu) = \frac{1}{\sigma\sqrt{2\pi}}$);
- σ 越小, 峰值越高(曲线越“瘦高”, 数据越集中); σ 越大, 峰值越低(曲线越“矮胖”, 数据越分散)(The smaller σ is, the higher the peak (the curve is “lean and tall”, and data is more concentrated); the larger σ is, the lower the peak (the curve is “short and fat”, and data is more scattered)).



3. **渐近性(Asymptotic Property):**

- 当 $|x| \rightarrow +\infty$ 时, $f(x) \rightarrow 0$ (曲线无限接近 x 轴)(As $|x| \rightarrow +\infty$, $f(x) \rightarrow 0$ (the curve approaches the x-axis infinitely));
- 实际意义(Practical Significance):正态变量取极端值的概率极低, 但并非为 0(The probability of a normal variable taking an extreme value is extremely low, but not zero).



4. **面积恒为 1(Constant Area of 1):**

- 曲线与 x 轴围成的面积为 1(总概率为 1)(The area under the curve and above the x-axis is 1 (total probability is 1));
- 区间概率 $P(a \leq X \leq b)$ 对应曲线在 $[a, b]$ 上的“曲边梯形面积”(The interval probability $P(a \leq X \leq b)$ corresponds to the “curvilinear trapezoid area” of the curve over $[a, b]$).

5. **拐点在 $x = \mu \pm \sigma$ 处(Inflection Points at $x = \mu \pm \sigma$):**

曲线在 $[\mu - \sigma, \mu + \sigma]$ 内向下凸 (concave downward), 在区间外向上凸 (convex upward);

这是 3σ 原则的几何基础, 体现了曲线“中间集中、两端平缓”的形态特征 (This is the geometric basis of the 3σ principle, reflecting the “concentrated in the middle and gentle at both ends” shape of the curve)。

第四章正态分布的数字特征

(Chapter 4 Numerical Characteristics of Normal Distribution)

正态分布的核心数字特征由 μ 和 σ 直接决定(The core numerical characteristics of a normal distribution are directly determined by μ and σ):

1. 数学期望(均值)(Mathematical Expectation/Mean): $E(X) = \mu$
2. 方差(Variance): $D(X) = \sigma^2$

4.1 样本均值与样本标准差的计算(Calculation of Sample Mean and Sample Standard Deviation)

实际应用中, 用**样本均值**(sample mean)估计 μ , 用**样本标准差**(sample standard deviation)估计 σ :

例(Example):抽取 5 名学生身高(cm):165、170、175、168、172, 计算样本均值和样本标准差(Calculate the sample mean and sample standard deviation)。

解(Solution):

- 样本均值(Sample Mean):

$$\bar{x} = \frac{165+170+175+168+172}{5} = 170 ;$$

- 样本标准差(Sample Standard Deviation):

$$s = \sqrt{\frac{(165-170)^2+(170-170)^2+(175-170)^2+(168-170)^2+(172-170)^2}{5-1}} \approx 3.808 。$$

第五章正态分布的概率计算:3σ原则

(Chapter 5 Probability Calculation of Normal Distribution: 3σ Principle)

对于 $X \sim N(\mu, \sigma^2)$, 随机变量落在 $\mu \pm k\sigma$ 区间内的概率仅与 k 有关, 这一规律称为 **3σ原则**(3σ principle)。

5.1 3σ原则的概率区间表(Probability Interval Table of 3σ Principle)

区间 (Interval)	概率值 (Probability)	剩余概率(单侧) (Residual Probability (One-sided))	实际意义(Practical Significance)
$[\mu - \sigma, \mu + \sigma]$	68.27%	15.865%	约 2/3 的数据集中在均值 $\pm 1\sigma$ 内 (About 2/3 of the data is concentrated within $\mu \pm 1\sigma$)
$[\mu - 2\sigma, \mu + 2\sigma]$	95.45%	2.275%	约 95%的数据集中在均值 $\pm 2\sigma$ 内 (About 95% of the data is concentrated within $\mu \pm 2\sigma$)
$[\mu - 3\sigma, \mu + 3\sigma]$	99.73%	0.135%	近 100%的数据集中在均值 $\pm 3\sigma$ 内 (Nearly 100% of the data is concentrated within $\mu \pm 3\sigma$)

5.2 概率计算的通用方法:标准化变换(General Method for Probability Calculation: Standardization Transformation)

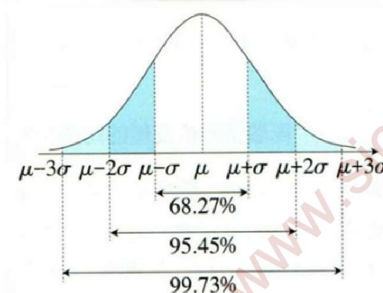
将一般正态分布转化为标准正态分布(standard normal distribution, $Z \sim N(0,1)$), 再查标准正态分布表(standard normal distribution table)计算概率(calculate probability):

$$Z = \frac{X - \mu}{\sigma} \Rightarrow P(a \leq X \leq b) = P\left(\frac{a - \mu}{\sigma} \leq Z \leq \frac{b - \mu}{\sigma}\right)$$

例(Example): 已知 $X \sim N(10, 4)$ ($\mu = 10, \sigma = 2$), 求 $P(7 \leq X \leq 13)$ (Find $P(7 \leq X \leq 13)$)。

解(Solution):

- 标准化(Standardization): $Z_1 = \frac{7-10}{2} = -1.5$, $Z_2 = \frac{13-10}{2} = 1.5$;
- 查标准正态分布表得(From the standard normal distribution table): $P(-1.5 \leq Z \leq 1.5) \approx 0.8664$, 即 $P(7 \leq X \leq 13) \approx 86.64\%$ (i.e., $P(7 \leq X \leq 13) \approx 86.64\%$)。



5.3 3σ原则的应用案例(Application Cases of 3σ Principle)

1. 质量控制(Quality Control):

某螺丝直径 $X \sim N(5, 0.04)$ (mm), 规定直径在 5 ± 0.6 mm 内为合格品(A certain screw's diameter $X \sim N(5, 0.04)$ (mm), and the diameter within 5 ± 0.6 mm is specified as qualified)。

解(Solution): $\sigma = 0.2$, $3\sigma = 0.6$, 不合格率仅 $1 - 99.73\% = 0.27\%$, 符合工业标准(The unqualified rate is only $1 - 99.73\% = 0.27\%$, which meets industrial standards)。

2. 考试分析(Exam Analysis):

高考数学成绩 $X \sim N(80, 100)$, 求成绩 >110 分的概率(The college entrance examination math score $X \sim N(80, 100)$, find the probability of score > 110)。

解(Solution): $110 = 80 + 3\sigma$, $P(X > 110) = \frac{1-99.73\%}{2} = 0.135\%$, 即仅 0.135%的考生成绩超 110 分

(i.e., only 0.135% of candidates have scores exceeding 110)。

第六章典型例题:正态分布的实际决策

(Chapter 6 Typical Example: Practical Decision-Making with Normal Distribution)

例(Example):

李明上学方式选择(Li Ming's Choice of Commuting Mode):坐公交用时(Time by Bus): $X \sim N(30, 6^2)$ (分钟, minutes); 骑车用时(Time by Bike): $Y \sim N(34, 2^2)$ (分钟, minutes)。

(1)若有 38 分钟可用, 选哪种方式? (If 38 minutes are available, which mode to choose?)

(2)若只有 34 分钟可用, 选哪种方式? (If only 34 minutes are available, which mode to choose?)

解题思路(Problem-Solving Ideas):

(1)38 分钟可用时(When 38 minutes are available):

公交(Bus): $38 - 30 = 8$, $8/6 \approx 1.33\sigma$, $P(Z \leq 1.33) \approx 0.9082$;

骑车(Bike): $38 - 34 = 4$, $4/2 = 2\sigma$, $P(Z \leq 2) = 0.9772$;

结论(Conclusion):选骑车(不迟到概率更高)

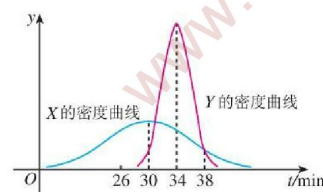
(Choose biking (higher probability of not being late))。

(2)34 分钟可用时(When only 34 minutes are available):

公交(Bus): $34 - 30 = 4$, $4/6 \approx 0.67\sigma$, $P(Z \leq 0.67) \approx 0.7486$;

骑车(Bike): $34 = \mu_Y$, $P(Y \leq 34) = 0.5$;

结论(Conclusion):选公交(不迟到概率更高)(Choose bus (higher probability of not being late))。



第七章正态分布的广泛应用

(Chapter 7 Wide Applications of Normal Distribution)

正态分布是统计学中应用最广的分布(The normal distribution is the most widely used distribution in statistics), 典型场景包括(typical scenarios include):

1. 医疗领域(Medical Field):

人体血压、血糖等生理指标服从正态分布, 医生通过“均值 $\pm 3\sigma$ ”判断指标是否异常(Physiological indicators such as human blood pressure and blood glucose follow a normal distribution, and doctors judge whether the indicators are abnormal through “mean $\pm 3\sigma$ ”)(如血糖 $N(5.1, 0.5^2)$, 若 $>6.6\text{mmol/L}$ 则异常)(e.g., blood glucose $N(5.1, 0.5^2)$, abnormal if $>6.6\text{mmol/L}$)。

2. 金融领域(Financial Field):

股票日收益率(剔除极端事件)近似服从正态分布, 金融机构用 3σ 原则计算“风险值(VaR)”, 评估极端损失概率(Daily stock returns (excluding extreme events) approximately follow a normal distribution, and financial institutions use the 3σ principle to calculate “Value at Risk (VaR)” and assess the probability of extreme losses)。

3. 教育测评(Educational Evaluation):

大规模考试成绩服从正态分布, 命题组通过 μ (平均分)调整试题难度, 通过 σ (区分度)判断试题是否能区分学生水平(Scores of large-scale exams follow a normal distribution. Question setters adjust the difficulty of questions through μ (average score) and judge whether questions can distinguish students' levels through σ (discrimination))。

第八章知识小结

(Chapter 8 Knowledge Summary)

1. 易混点(Confusing Points):

$N(\mu, \sigma^2)$ 的第二个参数是方差(variance), 不是标准差(standard deviation);

2. 易错点(Common Mistakes):

连续型随机变量 $P(X = x_0) = 0$, 但“ $X = x_0$ ”并非不可能事件(Continuous random variable $P(X = x_0) = 0$, but “ $X = x_0$ ” is not an impossible event);

3. 关键点(Key Points):

3σ 原则是“近似概率”(approximate probability), 正态变量仍可能超出 $\mu \pm 3\sigma$, 只是概率极低(normal variables may still exceed $\mu \pm 3\sigma$, but the probability is extremely low);

4. 实用技巧(Practical Skills):

概率计算优先用“对称性”(symmetry)和“标准化变换”(standardization transformation)简化运算(simplify calculations)。